IJTABA HASAN

Rawalpindi, Pakistan |03330643251 | iijtaba.hasan@gmail.com |www.linkedin.com/in/ijtaba-hasan-509b58308 | https://github.com/Ij-taba

PROFESSIONAL SUMMARY

AI and Machine Learning engineer with a BS in Artificial Intelligence (CGPA 3.82) and proven results across Computer Vision, NLP, and Generative AI — including a YOLO model trained on 50,000 images at 88% accuracy, an LSTM achieving 92% on duplicate question detection, and a BERT classifier fine-tuned to 90% across seven medical categories. Experienced building production-grade systems: RAG pipelines, automated n8n agents, and a full MLOps workflow with CI/CD on AWS. A published researcher who ships working models, not just prototypes.

PROFESSIONAL EXPERIENCE

ML Intern

Dione, Rawalpindi December 2023 – June 2024

- Contributed to computer vision projects including object detection and image classification using YOLO.
- Collected and organised a dataset of 50,000 drone images of roads and vehicles for model training.
- Annotated 50,000 drone images using Roboflow, ensuring accurate labelling and dataset quality for model training.
- Trained and optimised a YOLO model on 50,000 images, achieving 88% detection accuracy on the test set.
- Deployed the trained YOLO model via a Flask backend for real-time inference.

ML Intern

Ezitech, Rawalpindi August 2024 – November 2024

- Developed and evaluated end-to-end Machine Learning and Deep Learning models across classification and regression tasks.
- Applied data cleaning and preprocessing techniques to prepare high-quality datasets for model training.
- Trained an LSTM model on 20,000 samples for duplicate question analysis, achieving 92% accuracy on the test set
- Strengthened expertise in model architectures, neural network design, and performance evaluation methodologies
- Built a question-answering chatbot for Ezitech. BLEU Score: **0.78**

Agentic AI Intern

Devrolin, Rawalpindi *September* 2025 – October 2025

- Designed and deployed agentic workflows using n8n.
- Developed automation agents for lead generation and content generation
- Implemented workflows for data extraction, web scraping, and Google Sheets integration
- Integrated APIs and multi-step logic to deliver end-to-end automated pipelines
- Delivered agent-based systems and real-world workflow automation using n8n.

AI Engineer — Project-Based

Research-Oriented 2024 – Present

- Designed and implemented AI systems across NLP, Computer Vision, and LLM-based applications.
- Built end-to-end ML pipelines including preprocessing, training, evaluation, and API deployment using Flask and Fast API
- Developed production-style solutions such as n8n agents, RAG chatbots, and YOLO-based object detection.

TECHNICAL SKILLS

Deep Learning & Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, CNNs, RNNs, LSTMs

Computer Vision: YOLO, OpenCV, Image Annotation, Object Detection, Image Classification

Natural Language Processing: NLTK,SpaCy

Data Science: Python, Pandas, NumPy, Matplotlib, Seaborn, SQL

Generative AI & Agentic AI: LangChain, RAG (Retrieval-Augmented Generation), n8n, LangGraph , Transformers, Model Finetuning

App Development: React Native, Flask (Backend API Development), AI Model Integration (ML/DL models into mobile and web applications)

MLOps & Cloud: AWS(S3,EC2,ECR), Experiment Tracking,Data Validation,C1/CD(github action), Model Deployment

Tools & Technologies: Docker · Git · GitHub · Docker Hub · VS Code · Hugging Face

Programming Languages: Python, Java, JavaScript, SQL

PROJECTS

Meditranscribe(final year project)

- Developed an AI-powered medical application that transcribes doctor–patient conversations and converts them into structured medical prescriptions.
- Implemented speech-to-text processing and NLP-based segmentation to accurately extract and organize clinical information.
- Designed a secure role-based access system supporting four user roles: doctor, patient, nurse, and receptionist, improving hospital workflow efficiency.
- Built an end-to-end patient medical record management system that reduces manual documentation errors and streamlines operations.
- Collected 35,000 labeled samples across 7 classes and fine-tuned a BERT model for classification, achieving 90% accuracy.

Ezitech Chatbot GitHub

- Built a Retrieval-Augmented Generation (RAG) chatbot using company PDFs as the knowledge source for customer support and internal queries.
- Implemented text chunking, embedding generation, and FAISS-based vector storage for fast and accurate similarity search.
- Retrieved the most relevant context chunks for each user query and passed them to an LLM to generate precise, contextaware responses.
- Reduced hallucinations significantly and achieved response times under 5 seconds, demonstrating a scalable real-world generative AI solution.

Visa Approval System GitHub

- Developed and deployed an end-to-end **Visa Approval Prediction System** using FastAPI, accepting applicant parameters (continent, education, job experience, prevailing wage, employment region, etc.) and returning real-time approval predictions backed by a MongoDB database.
- Implemented a complete **MLOps pipeline** including data validation with Evidently AI, a scratch-built data version control system, experiment tracking, and automated CI/CD deployment to AWS using GitHub Actions.
- Engineered a **production-grade ML workflow** covering data ingestion, preprocessing, model training, evaluation, and monitoring — ensuring reproducibility, drift detection, and scalable cloud deployment on AWS with full pipeline automation.

License plate recognition system GitHub

- Developed a license plate recognition system by collecting and preparing a dataset of 20,000 vehicle number plate images.
- Trained a YOLO-based object detection model to detect license plates, achieving 75% detection accuracy.
- Implemented OpenCV-based post-processing to extract detected license plates for further analysis.
- Enabled real-time vehicle identification suitable for parking management, traffic monitoring, and security applications.

Appointment Agent (n8n)

- Developed a smart workflow automation agent using n8n to efficiently manage doctor appointments.
- Enabled patients to book and cancel appointments through a chat-based interface.
- Allowed doctors to view, track, and manage schedules and patient records in a centralized system.
- Automated notifications and appointment reminders to improve communication and reduce no-shows.
- Streamlined appointment management workflows, significantly reducing manual administrative effort and improving operational efficiency.

Content Generation Agent (n8n)

- Built an automated content generation agent using n8n
- Configured a 24-hour scheduled trigger to fetch topics from Google Sheets
- Generated structured prompts dynamically based on each topic
- Integrated the VEO3 API to automatically generate videos from prompts
- Stored generated video links back into Google Sheets for tracking and reuse

Duplicate Question Analysis GitHub

- Developed an LSTM-based NLP model to detect duplicate and semantically similar questions.
- Trained the model on a dataset of 20,000 samples, achieving 92% accuracy in duplicate question classification.
- Improved knowledge retrieval by reducing redundant entries in question–answer platforms.
- Enhanced information access and organization for educational and corporate datasets.

EDUCATION

BSCS (Artificial Intelligence) 2022-2026

BIIT PMAS Arid Agriculture University, Rawalpindi

CGPA: 3.82

Publications & Research

AI-Driven Doctor–Patient Dialogue Transcription and Automated Clinical Documentation System

Under Review /Accepted for Presentation at the *1st International Conference on AI and Multidisciplinary Innovations*